

ARTILLERY (1885–96)

In 1855, British engineer William Armstrong developed the first effective breech-loading, rifled field gun. While breech-loading was quicker than loading via the muzzle, rates of fire increased dramatically after 1885 with the introduction of integrated ammunition. This ammunition consisted of primer, propellant charge, and projectile, all contained in a brass cartridge case, was similar to the small-arms rounds developed only a few years before. Rapidly firing artillery using these new cartridges were described as “Quick-Fire” or “QF” guns. Other breech-loaders used no cartridge case—the explosion of the propellant was contained by a special seal, or obturator, on the breechblock. Projectiles fired by artillery pieces in the smoothbore era were spherical and had predictable weights. For example, a 6.4in caliber weapon always fired a 32lb (14.5kg) projectile and was called a “32-pounder.” With the coming of rifled artillery, projectiles could be made in a range of shapes and weights for a given caliber. Yet some weapons continued to be described in terms of the weight of the solid projectiles they would shoot if they were smoothbore.

► HOTCHKISS QF 3-POUNDER NAVAL GUN

Date	1885
Origin	France
Length (Barrel)	6½ft (2m)
Caliber	47mm
Range	2¼miles (3.6km)

The breech-loading Hotchkiss QF 3-pounder was used by the British Royal Navy from 1885, as well as the French, Russian, and US navies. These guns, made by a division of the Armstrong armaments business, were designed to fire at fast torpedo boats. Operated by two men, they could achieve a rate of fire of about 25 steel shells per minute, an incredibly high rate for the period.

Traversing ring

Naval pintle mounting

Steel barrel

Carriage wheel

► BREECH-LOADING 15-POUNDER 7CWT

Date	1892
Origin	UK
Length (Barrel)	7ft (2.13m)
Caliber	76.2mm
Range	3¼ miles (5.26km)

This light field gun could fire eight rounds per minute. It had a barrel weight of 7 cwt (7 hundredweight/784lb). It was equipped with an early recoil device—its spade was connected to a spring recoil buffer. When fired, the gun was aligned in such a way that the spade dug into the ground, compressing the spring. The elasticity of the spring stopped the rearward movement of the gun and pushed it back to its original position. Stability during operation meant that the gun fired its projectile at the intended angle, and the crew was not injured by the entire piece leaping backward.

FULL VIEW